

If the Amber Standby LED is Not Working

With the power turned off and the signal generator plugged in, the front panel amber standby LED should be on. The amber Standby LED is controlled by a +15 Vdc standby line supplied by the power supply, which is routed through the A23 Motherboard to the front panel.

To troubleshoot the +15 Vdc Standby line, turn the signal generator off and follow the signal path. Refer to [Table 1-12](#), [Table 1-13](#), [Table 1-14](#), and [Table 1-15](#).

If the Green Line Power LED is Not Working

When the power switch is turned on, the amber LED goes out and the green LED comes on. The fans start rotating and the front panel display becomes illuminated.

To troubleshoot, refer to [Table 1-12](#), [Table 1-13](#), [Table 1-14](#), and [Table 1-15](#) and verify the power supply voltages at J28 of the A23 Motherboard. If the supply voltages are correct, replace the A3 Front Panel Auxiliary Board. (Refer to “[A3 Front Panel Auxiliary Board](#)” on page 3-17.)

If the Fans are Not Working

There are three fans located near the front of the instrument that are connected to the A23 Motherboard. The fan control voltages are supplied by the A6 Power Supply and may be measured at J28 of the A23 Motherboard. The fan control voltage range is +8.5 Vdc (at 23 degrees C) to +12.0 Vdc (at 55 degrees C). At higher temperatures, the fan voltage increases along with the rotation of the fan.

To troubleshoot, disconnect each fan (one at a time) from the A23 Motherboard and check the fan voltage. Refer to [Table 1-12](#), [Table 1-13](#), [Table 1-14](#), and [Table 1-15](#). If the fan voltage is correct, replace the fan. (Refer to “[B1 Power Supply Fan](#)” on page 3-76, “[B2 Small Fan](#)” on page 3-78, or “[B3 Daughterboard/Card Cage Fan](#)” on page 3-80.)

If the Signal Generator Does Not Power-on and the Power Supply LEDs are Not Working

Each of the power supplies has an LED located on the bottom of the A23 Motherboard. If the power supply is functioning, the green LED is on. Use a voltmeter to measure the supplies on the A23 Motherboard to ensure the voltages meet the power supply specifications in the following table.

With the exceptions of the +15 Vdc and the +15 Vdc Standby power supply, each power supply should have a maximum of 10 mV_{p-p} ripple. The +15 Vdc and the +15 Vdc Standby power supply have a maximum ripple specification of 20 mV_{p-p}. The ripple can be measured using an oscilloscope. If any of the supplies LEDs are not on or the measured voltage is less than the acceptable value, one of the assemblies may be loading down the power supply. To find where each power supply is used, refer to [Table 1-12](#), [Table 1-13](#), [Table 1-14](#), and [Table 1-15](#).

To troubleshoot and isolate a defective assembly, turn power off to the signal generator and remove one of the assemblies being biased by the faulty power supply.

Power Supply	Acceptable Voltage
+32 Vdc	32.0 ± 0.96 Vdc
+15 Vdc	15.0 ± 0.45 Vdc
+15 Vdc Standby	15.0 ± 0.45 Vdc
+9.0 Vdc	9.0 ± 0.27 Vdc
+5.2 Vdc	5.2 ± 0.16 Vdc
+3.3 Vdc	3.3 ± 0.10 Vdc
−6.0 Vdc	−6.0 ± 0.18 Vdc
−15 Vdc	−15.0 ± 0.45 Vdc

NOTE

It is important to have a minimum number of assemblies removed at one time. The power supply expects to see a minimum load, otherwise the power supply voltage will increase until an over-voltage condition exists.

Turn power back on to the signal generator and check the suspected faulty power supply. If the condition is still bad, turn power off to the signal generator and re-install the first assembly that was removed. Remove one of the other assemblies being

biased by the faulty power supply and see if the problem is alleviated. Continue this process until the power supply is functioning properly. Replace the last assembly that was removed.

If the Front Panel Power Switch is Not Functioning

When the front-panel power switch is switched on, the control line (ON/OFF) from the A1 Keyboard is momentarily grounded and the A6 Power Supply is enabled. The ON/OFF signal is routed from the A1 Keyboard through J32-20 to the A23 Motherboard. Circuitry on the A23 Motherboard uses the ON/OFF signal to generate the Pwr_On_L. This line is routed to the A6 Power Supply and the A12 CPU; the A6 Power Supply is disabled when the Pwr_On_L line is a TTL high or open.

To troubleshoot

1. Measure J28-34 with a DVM to assure that the +15 Vdc Standby voltage (+15 STBY) is correct.
 - If the +15 Vdc Standby voltage is good, use the DVM to measure J32-20 (ON/OFF). The DVM should read approximately +5 Vdc.
 - a. While monitoring the DVM, depress and hold the front panel power switch. This grounds the ON/OFF signal and the DVM should read less than a few millivolts.
 - b. Release the front panel power switch. The ON/OFF control signal should again be approximately +5 Vdc.
2. With the front panel power switch off, the Pwr_On_L line at J28-32 of the A23 Motherboard should be at approximately +5.8 Vdc; this signal is generated on the A23 Motherboard.
 - If the voltage is less than +3.5 Vdc, remove the A6 Power Supply and measure the resistance of the connection from J28-32 to ground; the impedance should be ≥ 1 Megaohm.
 - If the impedance is correct, replace the A6 Power Supply.
3. When the front panel power switch is turned on, the Pwr_On_L line goes to $< +50$ mV.
 - If the A1 Keyboard is functioning properly and the Pwr_On_L line is not, replace the A23 Motherboard.
 - If the Pwr_On_L line is functioning properly, replace the A6 Power Supply.

2009 ALC Detector Test

- Replace the A14 Extended Frequency Output.

2010 Bulk R DAC Test

- Replace the A14 Extended Frequency Output.

2011 Log Offset Test

- Replace the A14 Extended Frequency Output.

Self-Tests 21xx for the A23 Motherboard

Figure 1-24

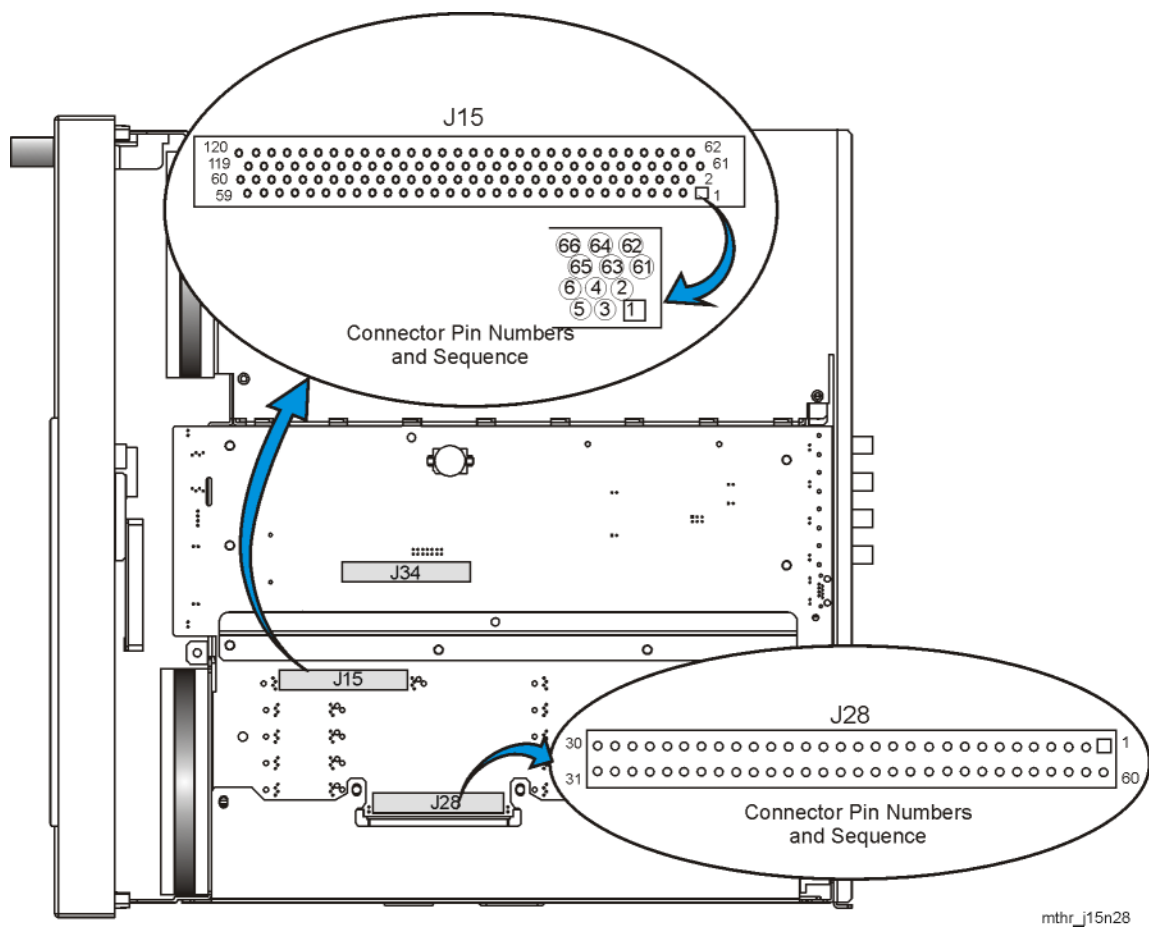


Table 1-30

Power Supply Voltage (Vdc)	Connector Pins on the A23 Motherboard	Minimum Value (Vdc)	Maximum Value (Vdc)	Origin
+32	J28-27	+31.04	+32.96	Main Supply
+15	J28-25, 26	+14.55	+15.45	Main Supply

Table 1-30

-15	J28-23	-15.45	-14.55	Main Supply
+9	J28-19, 42	+8.73	+9.27	Main Supply
-6	J28-38	-6.18	-5.82	Main Supply
-5.2	J28-21, 40	-5.356	-5.044	Main Supply
+5.2 VD	J28-14, 15, 46, 47	+5.044	+5.356	Main Supply
+3.4 VD	J28-1, 2, 5, 6, 9, 10, 51, 52, 55, 56, 59, 60	+3.298	+3.502	Main Supply

2100 Power Supply

- If the supply voltages are good on the connector pins indicated, replace the A23 Motherboard. (Refer to “[A23 Motherboard](#)” on page 3-56.)
- If the supply voltages are bad on the connector pins indicated:
 - a. Check the origin of the supply.
 - If the supply voltage is good at the origin, replace the A23 Motherboard. (Refer to “[A23 Motherboard](#)” on page 3-56.)
 - If the supply voltage is bad at the origin, troubleshoot the problem using that assembly's troubleshooting procedure.

2101 Sweep Diag

- Replace the A23 Motherboard.

2102 LCD Bias

- Use a DVM to probe A23J15-35 for a signal of approximately 21 Vdc.
 - If the voltage is good, replace the A23 Motherboard. (Refer to “[A23 Motherboard](#)” on page 3-56.)
 - If the voltage is bad, troubleshoot the A12 CPU.